CV-014

Staff Firmware Engineer

Infineon Technologies · India · Engineering

Skills

Embedded C, C/C++, Python, Assembly, FreeRTOS, Zephyr, RUST, AUTOSAR, ASPICE, MISRA-C, Polyspace, Trace32, UDE, gdb, Git, Jenkins, Make/CMake, Linux, Sparx Enterprise Architect, JIRA, Agile, Assembly (TriCore, x86)

Work experience

Staff Firmware Engineer · Infineon Technologies

2023-10 – Present

Developed AURIX TC4xx startup code (CRT0) for TriCore multi-core bring-up, including clock/PLL setup, memory initialization, stack/heap configuration, and interrupt vector tables.
Developed low-level device drivers for AES, RADAR, RDMA, SPU, AUDIO, PSI5, SENT, UART, I2C leveraging register-level programming and hardware abstraction layers (HAL).
Implemented security and cryptography algorithms such as AES128.
Testing the feature and driver for the owned IPs on Virtualizer Platform, Simulation, Emulation and Silicon.
Developed linker scripts across Tasking, HighTec, GCC, and GHS toolchains…

SoC Design Engineer · Intel

2021-04 – 2023-09

Developed Assembly firmware for Automatic Voltage Scaling (AVS) on Intel Rialto Bridge GPU and migrated firmware features from PVC to Rialto.
Designed stimulus testbenches for the Cluster Test Environment and Python test-suite generators to refine emulation-model validation.
Developed low-level C/Assembly drivers and test suites for emulation-model validation and improved emulation accuracy.

Expert Validation Engineer · Infineon Technologies

2018-07 – 2021-04

Validated firmware, reset, and clock behavior on Zebu/Veloce emulators and on silicon Automated clock validation using python to reduce efforts from 3 months to 3 days.
Developed and validated CAN-FD in Bootstrap Loader. Further advanced it to create a methodology using Vector and Python.
Applied machine learning to generate corner-case clock-ratio scenarios from existing datasets and used jitter-measurement algorithms to analyze multiplexed clocks.

Research Associate · Accelo Automoty India

2015-11 – 2017-01

Integrated Bosch BME280 and HC-06 Bluetooth modules with ATmega328P Developed a responsive website using WordPress, Bootstrap, HTML5, and CSS.

Education

- Yeshwantrao chavan college of engineering (2017)
- Associate's Degree — Shri Datta Meghe College of Engineering (2014)

Latest: Electrical Engineering — Yeshwantrao Chavan College Of Engineering